30/05/2025

Code-A_Phase-1

Aakash

Medical | IIT-JEE | Foundations

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MM : 240

TYM for NEET-2025-26_Practice Test-02A

Time : 60 Min.

PHYSICS

1. (2)
2. (4)
3. (2)
4. (1)
5. (3)
6. (3)
7. (1)
8. (3)
9. (1)
10. (3)
11. (4)
12. (4)
13. (1)
14. (2)
15. (3)

CHEMISTRY

16. (2)
17. (4)
18. (4)
19. (4)
20. (3)
21. (3)
22. (1)
23. (3)
24. (3)
25. (4)
26. (4)
27. (3)
28. (2)
29. (4)
30. (2)

BOTANY

31. (4)
32. (4)
33. (4)
34. (2)
35. (1)
36. (2)
39. (2)
40. (3)
41. (4)
42. (2)
43. (1)
44. (3)

37. (3)

38. (3)

45. (2)

ZOOLOGY

46. (2)

47. (4)

48. (4)

49. (3)

50. (3)

51. (1)

52. (4)

53. (2)

54. (3)

55. (4)

56. (3)

57. (2)

58. (4)

59. (4)

60. (2)

Hints and Solutions

PHYSICS

(1) Answer : (2)
Solution:
$y = x^2 - 4x$
$\frac{dy}{dx} = 2x - 4 = 0$
$x = 2$
$\frac{d^2y}{dx^2} = 2 > 0$
That means the minimum value of equation is at $x = 2$

(2) Answer : (4)
Solution:
Hint: $\therefore \int x^n dx = \frac{x^{n+1}}{n+1} + c$
Sol.: $\int_0^2 y dt = \int_0^2 (3t^2 + 2t)\, dt$
$= [t^3 + t^2]_0^2 = [2^3 - 0^3] + [2^2 - 0^2]$
$= 12$

(3) Answer : (2)
Solution:
Hint: $\frac{dy}{dx} = \frac{\left(\frac{dy}{d\theta}\right)}{\left(\frac{dx}{d\theta}\right)}$
Sol.: $y = \sin\theta$
$\frac{dy}{d\theta} = \cos\theta$
$x = \cos\theta$
$\frac{dx}{d\theta} = -\sin\theta$
$\frac{dy}{dx} = \frac{\cos\theta}{-\sin\theta}$
$\frac{dy}{dx} = -\cot\theta$
$\left(\frac{dt}{dx}\right)_{\theta = \pi/4} = -\cot\left(\frac{\pi}{4}\right) = -1$

(4) Answer : (1)
Solution:
Hint: Slope = $\frac{dy}{dx}$

Sol.: Slope = $\frac{d(x^3)}{dx} = 3x^2$
at $x = 1$
Slope = $3 \times (1)^2$
Slope = 3

(5) Answer : (3)
Solution:
Area bounded by curve y = f(x) and x-axis is
$A = \int_0^3 y dx = \int_0^3 3x^2 dx = 27\ m^2$

(6) Answer : (3)
Solution:
Slope = $\tan\theta$
θ_4 is maximum
$\therefore$ Slope for D is maximum

(7) Answer : (1)
Solution:
Slope $m = \tan 60°$
$= \sqrt{3}$
And intercept = – 5

(8) Answer : (3)
Solution:
$\int_0^3 t^2 dt = I$
$I = \left[\frac{t^3}{3}\right]_0^3$
$I = \frac{27}{3} = 9$

(9) Answer : (1)
Solution:
$y = \sin\theta + \sqrt{3}\cos\theta$
For maximum value of 'y'
$\frac{dy}{d\theta} = 0$
$\cos\theta - \sqrt{3}\sin\theta = 0$
$\cos\theta = \sqrt{3}\sin\theta$
$\tan\theta = \frac{1}{\sqrt{3}}$
$\therefore \theta = 30^\circ$
On applying this in y
$y_{\max} = \sin 30^\circ + \sqrt{3}\cos 30^\circ$
$= \frac{1}{2} + \sqrt{3}\left(\frac{\sqrt{3}}{2}\right)$
$= \frac{1}{2} + \frac{3}{2} = 2$

(10) Answer : (3)
Solution:
For maximum: $\frac{d^2y}{dx^2} < 0$
For minimum: $\frac{d^2y}{dx^2} > 0$

(11) Answer : (4)
Solution:
For *OA*; θ is acute so slope is +ve
For *AB*; θ is zero so slope is zero
For *BC*; θ is obtuse so slope is –ve
For *CD*; θ is increasing so slope is variable

(12) Answer : (4)
Solution:
$y = \sqrt{x} = x^{1/2}$
$\frac{dy}{dx} = \frac{1}{2}x^{\frac{1}{2}-1} = \frac{1}{2\sqrt{x}}$

(13) Answer : (1)
Solution:
$y = \sin 4x$
$\frac{dy}{dx} = \frac{d}{dx}\left(\sin 4x\right) \times \frac{d}{dx}\left(4x\right)$
$= 4\cos 4x$

(14) Answer : (2)
Solution:
$(x - x_1)^2 + (y - y_1)^2 = h^2$
Centre = (x_1, y_1)
= (–2, 1)

(15) Answer : (3)
Solution:

$S_n = 1 + 2 + 3 + \ldots + n$

$= \frac{n}{2}\left(1+n\right) = \frac{n(n+1)}{2}$

CHEMISTRY

(16) Answer : (2)

Solution:

$Zn + 2HCl \rightarrow ZnCl_2 + H_2(g)$

Mole of H_2 produced = $\frac{2.24}{22.4} = 0.1$

Mass of Zinc required = 65.4 × 0.1

= 6.54 g

(17) Answer : (4)

Solution:

Mole of $CH_4 = \frac{160}{16} = 10$

Mole of O_2 required = 10 × 2 = 20

Volume of O_2 required = 20 × 22.4 = 448 L

(18) Answer : (4)

Hint:

$\chi_{solute} = \frac{\text{moles of solute}}{\text{moles of solute+moles of solvent}}$

Solution:

Moles of NaOH = $\frac{4}{40} = 0.1$

Moles of H_2O = $\frac{36}{18} = 2$

So, $\chi_{NaOH} = \frac{0.1}{2.1} = \frac{1}{21}$, $\chi_{H_2O} = 1 - \frac{1}{21} = \frac{20}{21}$

(19) Answer : (4)

Hint:

	$2H_2(g)$	+ $O_2(g)$	$\rightarrow$ $2H_2O(\ell)$
(i)	5 mol	2 mol	–
(f)	1 mol	–	4 mol

∴ mass of H_2O formed = 4 × 18 = 72 g

(20) Answer : (3)

Solution:

Concentration term having volume in its expression is temperature dependent.

(21) Answer : (3)

Solution:

Number of gram equivalent of H_2SO_4 = Number of gram equivalent of NaOH solution

⇒ $N_1V_1 = N_2V_2$

⇒ $1 \times 2 \times V_1 = 20 \times 1$

⇒ $V_1 = 10$ ml

(22) Answer : (1)

Solution:

	$H_2(g)$ +	$Cl_2(g)$ →	$2HCl(g)$
	xmL	xmL	2xmL
	20 mL	20 mL	
After completion of reaction	–	–	40 mL

(23) Answer : (3)

Solution:

$\chi_{urea} = \frac{10/60}{\frac{10}{60}+\frac{90}{18}} = \frac{1/6}{\frac{1}{6}+5} = \frac{1}{31}$

(24) Answer : (3)

Solution:

$$\begin{array}{lll} N_2 & + \quad 3H_2 & \rightarrow \; 2NH_3 \\ 28\,g & \quad 6\,g & \quad 34\,g \\ & \quad 10\,g & \\ 56\,g & \lim iting & \\ & reagent & \end{array}$$

So, NH_3 formed $= \frac{34}{6} \times 10 = \frac{170}{3}\,g$ = 56.7 g

(25) Answer : (4)
Solution:

$C_2H_6 + \frac{7}{2}O_2 \rightarrow 2CO_2 + 3H_2O$

$n_{C_2H_6} = \frac{300}{30} = 10$ mol

1 mol C_2H_6 gives 2 mol CO_2

10 mol C_2H_6 gives 20 mol CO_2 = 20 × 44 g

= 880 g CO_2

(26) Answer : (4)
Solution:

Let the volume of solution be 1 L density = 1.02 g/mL

∴ Mass of solution = 1.02 × 1000 = 1020 g

∴ Mass of solute in 30% (w/w) aqueous solution = $\frac{1020 \times 30}{100} = 306$ g

∴ Molarity = $\frac{\frac{306}{166}}{1} = 1.84$ M

(27) Answer : (3)
Solution:

$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

Mole of CaO = $\frac{28}{56} = 0.5$

$CaCO_3$ = 0.5 mole

0.5 × 100 = 50 g

Quantity of $CaCO_3$ required = $\frac{100}{80} \times 50 = 62.5$ g

(28) Answer : (2)
Solution:

Number of glucose molecules = 200 × 0.4 × 10^{-3} × 6.02 × 10^{23} = 481.6 × 10^{20} = 4.8 × 10^{22}

(29) Answer : (4)
Solution:

Molality is the moles of solute dissolved per kg of solvent therefore 500 g, 1 molal solution contains 0.5 of solute, as

$$m = \frac{\text{Moles of solute}}{\text{Mass of solvent (in kg)}}$$

$$1 = \frac{0.5}{\text{Mass of solvent (in kg)}}$$

∴ Mass of solvent (in kg) = 0.5

= 500 g

(30) Answer : (2)
Hint:

$$\text{Molarity} = \frac{\text{number of moles of solute}}{\text{volume of solution in litre}}$$

Solution:

number of moles of solute $= \frac{12.04 \times 10^{22}}{6.02 \times 10^{23}}$

$= 2 \times 10^{-1}$

= 0.2 mole

$M = \frac{n}{V} = \frac{0.2}{500} \times 1000 = 0.4$ M

BOTANY

(31) Answer : (4)
Solution:
Cell wall is freely permeable to solutes.

(32) Answer : (4)
Solution:
Centriole has 9+0 arrangement of microtubules. The central part of centriole is made up of protein and is called hub.

(33) Answer : (4)
Solution:
Axoneme of flagella usually has nine doublets of radially arranged peripheral microtubules

(34) Answer : (2)
Solution:
The cytoskeleton in a cell is involved in many functions such as mechanical support, motility, maintenance of the shape of the cell.

(35) Answer : (1)
Solution:
Nucleolus is the site of the synthesis of rRNA.

(36) Answer : (2)
Solution:
Carotenoids containing coloured plastids are chromoplasts.

(37) Answer : (3)
Solution:
Mitochondrial matrix has 70S ribosomes.

(38) Answer : (3)
Solution:
The sap-vacuole is bound by a single membrane called tonoplast.

(39) Answer : (2)
Solution:
Answer (2)
Endomembrane system does not include mitochondria and chloroplast.

(40) Answer : (3)
Hint:
This organelle also facilitates excretion in *Amoeba.*
Solution:
In *Amoeba*, the contractile vacuole is important for the osmoregulation and excretion.

(41) Answer : (4)
Solution:
Golgi apparatus is the important site of formation of glycoproteins and glycolipids.

(42) Answer : (2)
Solution:
Robert Brown discovered nucleus in 1831.

(43) Answer : (1)
Solution:
Elaioplasts stores fats and oils.

(44) Answer : (3)
Solution:
Infoldings of inner membrane of mitochondria are called cristae.

(45) Answer : (2)
Solution:
Thylakoid stack to form grana.

ZOOLOGY

(46) Answer : (2)
Solution:
Shortening and lengthening of the muscle cell occurs due to contractility property.
Nerve fibres do not show contractility. Insulation in nerve fibres is provided by myelin sheath.

(47) Answer : (4)
Solution:
Cardiac muscle fibres are uninucleated.

(48) Answer : (4)
Solution:
Muscle fibres associated with biceps are voluntary and striated.

(49) Answer : (3)
Solution:
Smooth muscle fibres or cells are elongated and spindle-shaped *i.e.*, pointed or taper at the ends (fusiform) and are uninucleated.

(50) Answer : (3)
Hint:
Brain send the signals for locomotion.
Solution:
Muscles play an active role in all the movements of the body. Adipose tissues are specialised to store fats. Epithelial lining performs functions like absorption, secretion, protection, *etc.*

(51) Answer : (1)
Solution:
Neurons show excitability and conductivity. Extensibility is a feature of muscle fibres.

(52) Answer : (4)
Solution:
In neural tissue, the axon carries impulses away from cyton and dendrites carry impulses towards the cyton.

(53) Answer : (2)
Hint:
The compound is substituted with chlorine.
Solution:
To analyse chemical composition of a living tissue it is taken and grinded in trichloroacetic acid (Cl_3CCOOH) using a mortar and a pestle. Acetic acid, sodium chloride and palmitic acid are not used for grinding the living tissue to analyse their chemical composition.

(54) Answer : (3)
Solution:
Primary metabolites take part in normal physiological processes.

(55) Answer : (4)
Solution:
Polysaccharides are present in plant and fungal cell wall and exoskeleton of arthropods.

(56) Answer : (3)
Solution:

Component	% of the total cellular mass
Water	70-90
Proteins	10-15
Carbohydrates	3
Lipids	2
Nucleic acids	5-7
Ions	1

(57) Answer : (2)
Hint:

Fruit sugar
Solution:
Inulin is a polymer of fructose. It is a storage polysaccharide in roots and tubers of dahlia and related plants.
Starch is a polymer of glucose. Glucose is a hexose monosaccharide.
Sucrose is a disaccharide containing one glucose molecule and one fructose molecule.

(58) Answer : (4)
Solution:
Morphine → Alkaloid
Concanavalin A → Lectins
Vinblastine → Drug

(59) Answer : (4)
Solution:
The molecular weight of lipids does not exceed 800 Da.

(60) Answer : (2)
Hint:
This polysaccharide is also present in plant's cell wall.
Solution:
Plant's cell walls, cotton fibre and paper from plant pulp are cellulosic in nature.
Starch is present as store house of energy in plant tissues.
Inulin is a storage polysaccharide of roots and tubers of dahlia and related plants.